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Approximation Algorithms and Approximation Ratios: Introduction

Jobshop Minimum Makespan Scheduling

Video: Introduction to Jobshop Scheduling and Algorithm Design
24 min

Video: Analysis of Jobshop Scheduling
26 min

Lab: Interactive Notes: Jobshop Scheduling
2h

Quiz: Job Shop Scheduling Questions Submitted

Vertex Cover

Maximum Satisfiability Problem

Problem Set # 3

Congratulations! You passed!

Job Shop Scheduling Questions

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

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Due Feb 25, 11:59 PM IST

Receive grade

To Pass 80% or higher

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The questions below concern the following instance of the jobshop scheduling problem.

Job ID	Time
1	5
2	6
3	2
4	5
5	2
6	1
7	3
8	4

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1. Consider an instance of the minimum makespan scheduling above. Suppose we have three processors. What is the makespan achieved if we ran the greedy algorithm. You can simply use a pencil and paper to run the greedy algorithm to work out the answer. Assume that the algorithm processes the jobs in the increasing order of Job ID. **2 / 2 points**

11

Correct
Processor # 1: Job IDs 1, 5, Processor #2 : Job IDs: 2, 6, 7, Processor # 3: Job IDs 3, 4, 8 yields a makespace of 11 (processor # 3).

2. What is the makespan of the solution **1 / 1 point**

Processor	Job IDs
1	2, 3, 5
2	1, 6, 7
3	4, 8

10

Correct
Processor 1 achieves the makespan with time of 10

3. Suppose we sorted the jobs in decreasing order of their times instead and ran the greedy algorithm. What is the makespan of the optimal solution obtained? **2 / 2 points**

10

Correct
In fact it is the schedule above.

4. Without solving the optimal problem, we wish to find bounds on the optimal solution anyways. Which of the following options are valid arguments that serve to place bounds on the optimal solution. **6 / 6 points**

- [x] The following assignment:

Processor	Job IDs
1	2, 3, 5
2	1, 6, 7
3	4, 8

has a makespan of 10. And therefore, the optimal solution must have value ≤ 10.

Correct
Correct. In fact, you can combine it with a previous result and establish that this solution is actually optimal.

- [ ] The approximation algorithm was designed by Ronald L. Graham, a very famous computer scientist. It must be a great algorithm and thus the optimal solution must be at least 11.
- [x] The total times for all jobs is 28. No matter how we distribute it among three processors, at least one processor must have finish time more than $\frac{28}{3}$ or $9.3333\ldots$. Therefore, the optimal answer cannot be smaller than 10.

Correct
Correct!!

- [x] By pigeon hole principle, if we consider the first four job IDs 1, 2, 3, 4; at least two of the jobs must share the same processor in any assignment. This yields a lower bound of 7 on the optimal solution.

Correct
This argument is correct as well.

- [x] The algorithm is known to have an approximation ratio of $2 - \frac{1}{3} = \frac{5}{3}$. Therefore, the optimal solution has to be at least $11 \times \frac{3}{5} \approx 6.6$

Correct
Correct. In fact, the optimal solution must be at least 7 since we know that it has to be an integer (all job times are integers).